Graph-Based Molecular Generation via Adversarial Learning with Hierarchical Discriminators and Reward-Driven Objectives

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***Abstract*—Recent advances in deep-generative models have allowed the generation of chemically valid and novel molecules with desirable properties for further investigation. In this work, we provide such a model, integrating generative adversarial networks (GANs), hierarchical graph discrimination, and reinforcement learning. Our graph-based generator generates molecular graphs in the form of adjacency and node feature matrices. To improve robustness and stability during training, we implement a hierarchical differentiable pooling (DIFFPOOL)-based discriminator. This type of discriminator allows our model to learn coarse-grained graph representations during training. Additionally, we introduce a reinforcement learning component in which a policy network receives reward signals based on predicted chemical properties, guiding the generator toward structures with high drug-likeness and other target metrics. In evaluating our approach, we look for validity, uniqueness, novelty, and alignment with drug-likeness metric scores. The results demonstrate improved generation quality and training stability compared to baseline graph GANs, particularly when incorporating DIFFPOOL and reinforcement-guided reward shaping. Our findings suggest that the combination of hierarchical discrimination with reinforcement learning shows a positive trajectory for property-driven molecular design in drug discovery and materials science.**

***Index Terms*—GAN, Hierarchical discriminator, DIFFPOOL, Reinforcement learning, Validity, Uniqueness, Novelty, Drug-likeness**

Nomenclature

$A \in \mathbb{R}^{nxn}$	adjacency matrix
$B \in \mathbb{Z}^{nxn}$	bond order matrix (optional)
$G = (V, E)$	molecular graph with vertices V and edges E
$z \in \mathbb{R}^{d}$	latent vector
n	number of nodes (atoms)
f	number of node features (atom types)
$A \in [0, 1]^{n\times n}$	soft adjacency matrix
$T \in \mathbb{R}^{n\times f}$	node feature matrix (softmax over atom types)
$T_i \in \mathbb{R}^{f}$	one-hot (or softmax) encoded atomic type of node v_i
$\mathcal{A}(z)$	flattened adjacency matrix
$\mathcal{T}(z)$	flattened node features output
$\mathcal{H} = v_i \in V : t_i = 1$	set of hydrogen atoms
$R(M)$	total reward for generated molecule M
$R \leftarrow \alpha R + (1-\alpha)R(M)$, $\alpha = 0.99$	exponentially smoothed baseline reward
$\lambda_G = 3.0$, $\lambda_R = 7.0$	scaling factors for generator and reward loss in scaled loss computation
$x^{(0)} = T$	input node features
$\sigma(\cdot)$	activation function
$\hat{y} \in \mathbb{R}^{n\times 1}$	score per node critic output

I. Introduction

In our research, we used the QM9 dataset to train a GAN generator working alongside a hierarchical discriminator to generate chemically valid and novel molecules for scientific study. We used two different versions of the hierarchical discriminator, one with DIFFPOOL, and one without. In association with the generator and discriminator, we also implement a reward policy network to encourage the model's generation. The dataset we used, QM9, contained the following variables: mol_id, smiles, A, B, C, mu, alpha, homo, lumo, gap, r2, zpve, u0, u298, h298, g298, cv, u0_atom, u298_atom, h298_atom, g298_atom, Morgan_fingerprint, adj_matrix, node_features, and node_count. The variables that are significant for this work are adj_matrix, node_features, and node_count. These variables define the molecules that are fed into the generator during training. The adjacency matrix, adj_matrix, contains information on which bonds are present inside the molecule. The node features, node_features, contain information on which atoms are present within the molecule. In this study, we analyze the validity, novelty, uniqueness, and drug-likeness of molecules produced by the generator. The validity is based on whether or not the generated molecule is chemically valid, uniqueness is based on similarity of the generated molecule to previous molecules, novelty is determined by whether or not a similar molecule is already present within the training data, and drug-likeness is concerned with the similarity of the molecule generated to common drug-like molecules.

II. Literary Review

In recent years, AI-driven methods have gained traction in *de novo drug design, offering the potential to generate novel chemical compounds without reliance on predefined molecular databases. These methods offer the potential to eliminate the dependence on existing compound databases and accelerate the drug discovery pipeline with optimized pharmacological properties.*

Among the most promising machine learning tools applied to molecular generation are graph-based generative models. Unlike traditional representations such as SMILES (Simplified Molecular Input Line Entry System) strings, graph-based models treat molecules as undirected graphs, where atoms are represented as nodes and chemical bonds as edges. This approach more accurately captures molecular structure and allows for more chemically valid generation processes. Within this domain, Graph Neural Networks (GNNs) and Generative Adversarial Networks (GANs) have become foundational model architectures.

MolGAN, introduced by De Cao and Kipf[1], represents one of the earliest applications of GANs to molecular graph generation. It combines an implicit generative model with a reinforcement learning objective to generate molecular graphs that are both valid and optimized for specific chemical properties. The model eliminates the need for graph matching or predefined node orders by using an adjacency matrix representation of molecules. Despite demonstrating high chemical validity, MolGAN suffers from limited output diversity due to mode collapse—a common issue in GAN-based models—and lacks structural regularization mechanisms.

To address these limitations, Guarino et al. [2] proposed DiPol-GAN, an extension of MolGAN that introduces two key innovations. First, it incorporates Differentiable Pooling (DIFFPOOL), a hierarchical pooling mechanism that enables the network to learn coarse-grained representations of molecular graphs. This facilitates hierarchical feature extraction and better captures the global context of molecular structures. Second, the model integrates a reinforcement learning policy gradient framework to further optimize molecular generation based on property-specific reward functions. DiPol-GAN has shown improvements in the generation of property-compliant and structurally diverse molecules, however, it also introduces additional complexity in training and higher computational overhead.

Several alternative architectures have also been explored in the literature. For instance, GraphAF, considered by Shi et al.[6], employs autoregressive flow-based modeling to sequentially construct molecular graphs in a chemically valid manner. While this ensures structural integrity during generation, the method can be computationally intensive and less suited to multi-objective optimization. Similarly, latent variable models such as GraphDF, authored by Luo et. Al[7], attempt to decouple the representation and property optimization process, offering greater interpretability and flexibility, though often at the cost of sample efficiency.

Despite these advancements, open challenges persist. The generation of molecules that are simultaneously valid, diverse, and property-optimized remains non-trivial. Many models struggle with achieving all three objectives concurrently. Additionally, evaluation metrics in molecular generation are often inconsistent, making it difficult to directly compare model performance across studies. Reinforcement learning-based optimization strategies, while effective in fine-tuning molecular properties, can further complicate training stability and scalability.

In this context, the present study revisits the DiPol-GAN architecture to evaluate its practical utility under modern evaluation standards and datasets. By reproducing the original model and conducting systematic benchmarking, we aim to assess its ability to generate chemically valid and functionally relevant molecular graphs. Furthermore, we explore potential enhancements to DiPol-GAN's training protocol and reward shaping functions to improve output diversity and property alignment. This work contributes toward validating and extending graph-based generative approaches in de novo drug discovery.

III. Methods

IV. Equations

A. Adjacency Matrix to Molecule

1) Vertex Set (Atoms): Each atom is represented by a vertex:

$$V = \{v_i = Atom(t_i) | i = 1, 2, ..., n\} \tag{1}$$

Optionally, if chirality tags are given:

$$Chiral(v_i) = c_i \; for \; i \in C \subseteq \{1, ..., n\} \tag{2}$$

2) Edge Set (Bonds): For all i < j:

$$\text{If } A_{ij} > \tau :$$
$$o_{ij} = \begin{cases} B_{ij}, & \text{if } B_{ij} > 0 \\ 1, & \text{otherwise} \end{cases}$$
$$E \ni e_{ij} = (v_i, v_j, \beta(o_{ij}))$$

Where $\beta(o)$ maps bond order to bond type:

$$\beta(o) = \begin{cases} \text{SINGLE}, & o = 1 \\ \text{DOUBLE}, & o = 2 \\ \text{TRIPLE}, & o = 3 \\ \text{AROMATIC}, & o = 4 \\ \text{undefined}, & \text{otherwise} \end{cases} \tag{3}$$

3) Isolated Hydrogen Handling: Let:

$$H = i|t_i = 1 \wedge deg(v_i) = 0 \tag{4}$$

For each h $\in$ H:

$$j^* = \arg\max_{\substack{j \notin H \\ j \neq h}} A_{hj} \tag{5}$$

If $deg(v_{j^*}) <$ Valence$_{max} t_{j^*}$, then:

$$E \ni e_{hj^*} = (v_h, v_{j^*}, \text{SINGLE}) \tag{6}$$

Else:

$$V \leftarrow V \setminus \{v_h\} \tag{7}$$

4) Molecule: The final molecule is:

$$M = \text{Sanitize}(G = (V,\ E)) \tag{8}$$

B. Reward Policy Network

1) Validity Reward: Let M be a candidate molecule derived from (A, T). Let $\text{Valence}_{\max}(t_i)$ be the maximum valence for atom type t_i, and let $\deg(v_i)$ be the current explicit valence (degree) of atom $v_i \in V$.

$$R_{\text{validity}}(M) = \begin{cases} -0.5, & \text{if } M \text{ is None or fails sanitization} \\ \max\left(0,\ 5 - \left|\sum_{v_i \in V} (\text{Valence}_{\max}(t_i) - \deg(v_i))\right|\right), & \text{otherwise} \end{cases} \tag{9}$$

2) Stray Hydrogen Reward: Let $H \subset V$ be the set of hydrogen atoms, and let deg(v_i) be the degree of atom v_i.

$$R_{\text{stray}}(M) = \begin{cases} -0.5, & \text{if } M \text{ is None or } \exists v_i \in H \text{ such that } \deg(v_i) = 0 \\ 0.5, & \text{otherwise} \end{cases} \tag{10}$$

3) Uniqueness Reward: Let $\mathcal{G} = \{S_1, S_2, ..., S_n\}$ be the list of SMILES strings for previously generated molecules, and let $S(M)$ be the SMILES of the current molecule M. Let $count_{\mathcal{G}}(S(M))$ be the number of times $S(M)$ appears in $\mathcal{G}$.

$$R_{\text{unique}}(M, \mathcal{G}) = \begin{cases} -0.3, & \text{if } R_{\text{validity}}(M) = 0 \\ \max\left(0, \min\left(1, 1 - \frac{|G|+1}{\text{count}_G(S(M))}\right)\right), & \text{otherwise} \end{cases} \tag{11}$$

4) Novelty Reward: Let $\mathcal{T} \subset \mathbb{S}$ be the set of training SMILES strings.

$$R_{\text{novelty}}(M, T, \mathcal{G}) = \begin{cases} -0.3, & \text{if } R_{\text{validity}}(M) = 0 \\ -2.5, & \text{if } R_{\text{unique}}(M, \mathcal{G}) < 1 \\ -0.7, & \text{if } S(M) \in T \\ 2.3, & \text{otherwise} \end{cases} \tag{12}$$

5) Drug-likeness Reward: Let $Q(M) \in [0, 1]$ be the RDKit QED score of molecule M.

$$R_{\text{drug}}(M) = \begin{cases} Q(M), & \text{if } R_{\text{validity}}(M) > 0 \\ -0.3, & \text{otherwise} \end{cases} \tag{13}$$

6) Total Reward:

$$R_{\text{validity}}(M) + R_{\text{stray}}(M) + R_{\text{unique}}(M, \mathcal{G}) + R_{\text{novelty}}(M, \mathcal{T}, \mathbb{G}) + R_{\text{drug}}(M) \tag{14}$$

C. Generator

1) Generator Function: The generator is a function:

$$G\colon \mathbb{R}^d \to ([0,1]^{n \times n}, \mathbb{R}^{n \times f}) \tag{15}$$

constructed from a multi-layer perceptron (MLP):

$$MLP(z) = \mathcal{A}(Z) \parallel \mathcal{T}(z) \tag{16}$$

Where:

- $\mathcal{A}(z) \in \mathbb{R}^{n^2}$, reshaped and symmetrized:

$$A = \frac{1}{2}(\text{reshape})(\mathcal{A}(z)) + (\text{reshape}(\mathcal{A}(z))^T) \tag{17}$$

- Then row-wise softmax is applied to normalize:

$$A_{ij} = \frac{exp(A_{ij})}{\sum_k exp(A_{ik})} \tag{18}$$

- T is obtained from:

$$T = \text{reshape}(\mathcal{T}(z)) \in \mathbb{R}^{n \times f}, T_i = \text{softmax}(T_i) \tag{19}$$

- Then the atom types $t_i \in 1, 6, 7, 8, 9$ are selected by:

$$t_i = \beta(\arg\max_k T_{ik}) \tag{20}$$

  Where $\beta(o)$ is the bond type mapping (i.e., atomic number index map).

2) Hydrogen Fixing Logic: For each $v_i \in \mathcal{H}$, if:

$$\deg(v_i) = \sum_j A_{ij} < \tau \text{with } \tau = 0.5 \tag{21}$$

Then set:

$$A_{ij^*} = A_{j^*i} = 0.9 \text{ where } j^* = \arg\max_{j \neq i} A_{ij} \tag{22}$$

3) Critic Loss: Let D be the discriminator.
For real and generated pairs:

- $D(A, T) \in [0, 1]$
- Use smoothed labels: $y_{real} = 0.9$, $y_{fake} = 0.1$

Binary cross-entropy:

$$\mathcal{L}_D = \text{BCE}(y_{real}, D(A_{real}, T_{real})) + \text{BCE}(y_{fake}, D(A_{gen}, T_{gen})) \tag{23}$$

4) Generator Loss with Reinforcement Learning: Advantage:

$$A(M) = R(M) - \bar{R} \tag{24}$$

Generator loss:

$$\mathcal{L}_G = \text{BCE}(1, D(A_{gen}, T_{gen})) \tag{25}$$

Combined scaled loss:

$$\mathcal{L}_{scaled} = \lambda_G \times \mathcal{L}_G - \lambda_R \times \mathbb{E}_{z \sim \mathcal{N}(0,1)}[log(P(z)) \times A(M)] \tag{26}$$

D. Critic without Differentiable Pooling

1) First Layer (Dense Transformation):

$$x^{(1)} = \text{ReLU}(W_1 x^{(0)} + b_1),\ W_1 \in \mathbb{R}^{f \times 64} \tag{27}$$

2) Second Layer (Dense Transformation):

$$x^{(2)} = \text{ReLU}(W_2 x^{(1)} + b_2),\ W_2 \in \mathbb{R}^{64 \times 128} \tag{28}$$

3) Final Layer (Output Scores):

$$\hat{y} = \sigma(W_3 x^{(2)} + b_3),\ W_3 \in \mathbb{R}^{128 \times 1} \tag{29}$$

E. Critic with Differentiable Pooling

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